

1 Chloroplasts are involved in both the light-dependent reactions and light-independent reactions of photosynthesis.

(a) Which row of the table shows where the light-dependent reactions and light-independent reactions take place?

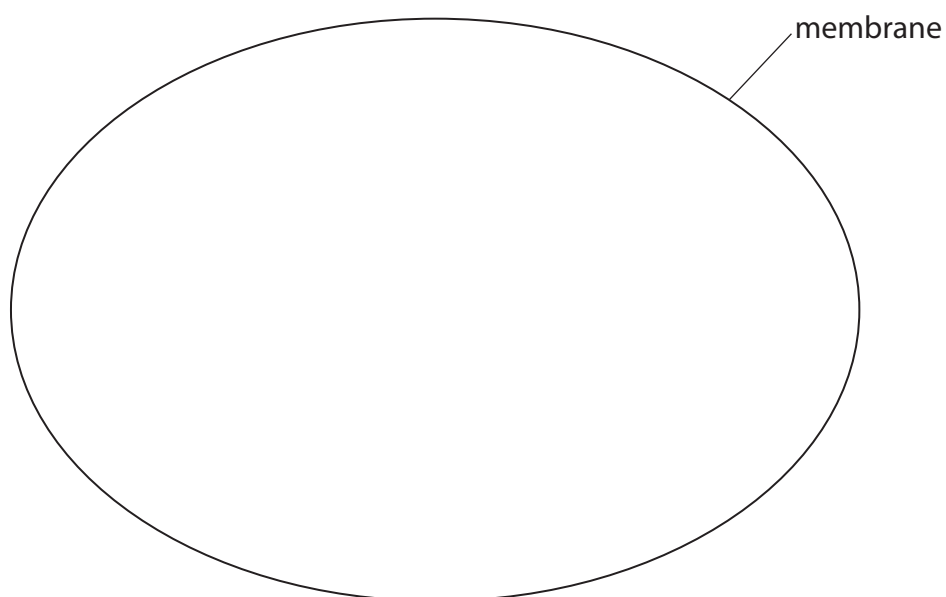
(1)

|                            | Light-dependent reactions | Light-independent reactions |
|----------------------------|---------------------------|-----------------------------|
| <input type="checkbox"/> A | stroma                    | stroma                      |
| <input type="checkbox"/> B | stroma                    | thylakoid membranes         |
| <input type="checkbox"/> C | thylakoid membranes       | stroma                      |
| <input type="checkbox"/> D | thylakoid membranes       | thylakoid membranes         |

(b) The diagram shows the outline of a chloroplast.

Draw **three** labelled features on this diagram that are found in a chloroplast, other than the stroma and the thylakoid membranes.

(3)



- (c) An absorption spectrum shows how much light is absorbed by chloroplasts at different wavelengths of light.

The table shows the colour of light at four wavelengths.

| Wavelength of light / nm | 460  | 520   | 600    | 680 |
|--------------------------|------|-------|--------|-----|
| Colour of light          | blue | green | yellow | red |

Which wavelength of light is absorbed the **least** by chloroplasts?

(1)

- ☐ A 460 nm
- ☐ B 520 nm
- ☐ C 600 nm
- ☐ D 680 nm

- (d) State what is meant by the term **action spectrum**.

(1)

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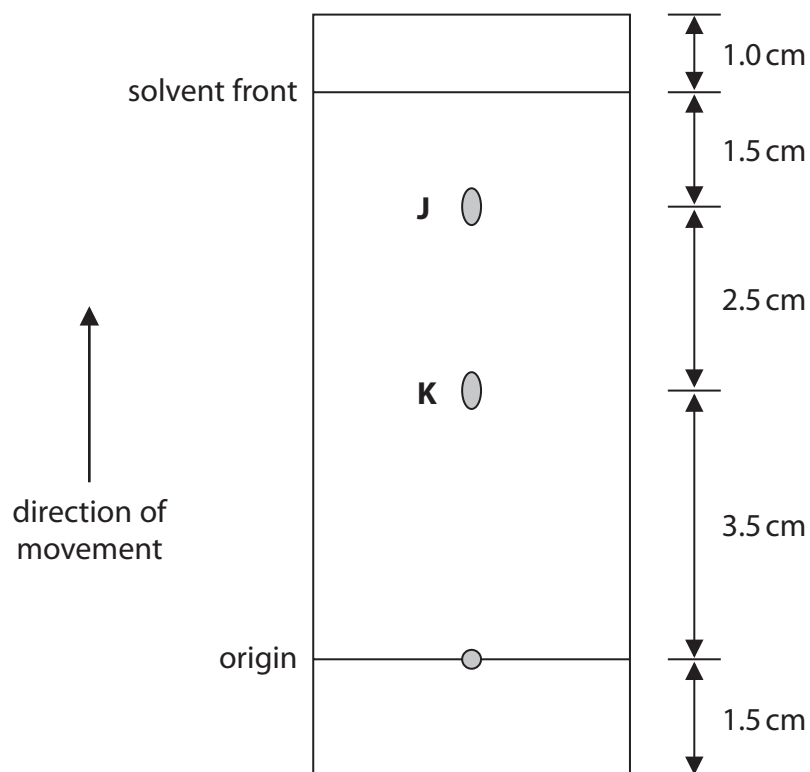
- (e) Chloroplast pigments can be separated and then identified by their  $R_f$  values.

- (i) Which process can be used to separate chloroplast pigments?

(1)

- ☐ A chromatography
- ☐ B dendrochronology
- ☐ C osmosis
- ☐ D PCR

(ii) The diagram shows separated chloroplast pigments, **J** and **K**.



What is the  $R_f$  value for chloroplast pigment **J**?

(1)

- ☐ **A** 0.625
- ☐ **B** 0.800
- ☐ **C** 0.830
- ☐ **D** 1.714

(Total for Question 1 = 8 marks)